

REDROB · INDIA RUNS DATA & AI CHALLENGE

HireFit Ranker

Ranks careers, not keywords.

A fast, offline, deterministic engine that surfaces genuinely hireable engineers from a 100,000-candidate pool — a shortlist a recruiter can trust.

github.com/bansalbhunesh/redrob-hirefit-ranker · live demo: redrob-hirefit-ranker.onrender.com



THE PROBLEM

The dataset punishes keyword filters

The JD says "5–9 yrs, embeddings/retrieval/ranking" — but it MEANS: find engineers who actually shipped ranking / search / recsys at product companies, even without the buzzwords. And reject:

Keyword stuffers

wrong-role profiles padded with AI skills

Honeypots

impossible profiles — "expert" in 10 skills with 0 months used

Unavailable

perfect on paper, but 5% response rate / not open to work

→ **So the ranker reads CAREERS and BEHAVIOR, not skill lists.**

THE APPROACH

Hybrid scoring + multiplicative guardrails



Final score = base_score × behavioral × honeypot × disqualifier

- BM25 lexical relevance + a 28-feature recruiter matrix (skills, career evidence, seniority, behavior, logistics).
- Guardrails are MULTIPLICATIVE — a high fit score cannot rescue an impossible, off-target, or unavailable profile.
- Career-evidence signals (production, IR/ranking) outweigh skill-list matches → Tier-5s without buzzwords still surface.
- Fully deterministic & offline: same input → byte-identical output; no network, no GPU, no LLM at rank time.

WHY THESE CHOICES

Deliberate, measured engineering

Feature matrix + BM25 — not an LLM per candidate

LLM-per-candidate can't scale to 100K on budget, isn't reproducible, needs network. We rank the pool on CPU in ~80–180s.

No dense embeddings — tested & rejected

Built a model2vec/potion branch, gated on a measured A/B: NDCG@10 +0.0000, ~2.2× runtime → FAIL. Shipped the simpler system.

Career-evidence over keywords

Production / IR-ranking signals from career history outweigh skill lists — how non-buzzword Tier-5s surface.

Multiplicative behavioral & honeypot guardrails

Encodes the JD's explicit "down-weight the unavailable" rule; impossible profiles forced to zero.

Deterministic everything

Byte-identical, traceable, debuggable — the basis for explainability and reproduction.

RESULTS

A shortlist a recruiter can trust

~104s

100K run · CPU only

100% offline

no network · GPU · LLM

0 / 53

honeypots in top output

P@10 · 1.0

independent LLM-judge

R

Redrob HireFit Ranker

v3.0

Showpiece

Live Proof

Batch

100,000

TOTAL CANDIDATES

100

RANKED OUTPUT

53

HONEYPOTS BLOCKED

0.722

AVG SCORE

PIPELINE ARCHITECTURE

Load

100,000

Text

100,000

BM25

100,000

28-D Features

100,000

Honeypot

53

Behavioral

100,000

Rank

100

Reasoning

100

All Candidates

Honeypots

Product Companies

Strong ML

Search by name, title, company...

1

Senior Machine Learning Engineer @ Zomato

CAND_0018499 | 7.2 YOE | Noida, Uttar Pradesh

1.0000

SCORE

Live interactive dashboard — pipeline stages, honeypot blocking, per-candidate audit.

VALIDATION

Proven without the hidden labels

1. Non-circular heuristic eval — an independent labeler sharing NO code with the ranker → composite 0.881 (rules out self-grading).
2. LLM-as-judge (dev-only, never in the ranking path) scored the top-10 against the JD:

top-10 tiers [5,5,4,4,5,5,5,5,5,5] · P@10 = 1.0 · NDCG@10 = 0.894

The judge even confirmed our behavioral guardrails are MORE recruiter-aware than itself — we correctly down-weighted high-skill candidates with 12% response rate that the LLM over-rated.

"We don't rank resumes. We rank hireable candidates."

EXPLAINABILITY

Every score is traceable — nothing is a black box

- Per-candidate REASONING in the output, generated only from facts in the profile — no hallucinated skills.
- Every score DECOMPOSES into named features + the three multipliers.
- Interactive dashboard shows the pipeline, honeypot blocking, and a live feature + reasoning audit.
- Two live UIs: Render dashboard + HuggingFace sandbox.

Ranked shortlist

rank	candidate_id	score	tier	reasoning
1	CAND_0000031	1.000	Top pick	Currently a Recommendation Systems Engineer at Swiggy with 6.0 years; aligns with the JD requirement for shipped retrieval/ranking systems; relevant skills include Embeddings, Pinecone, scikit-learn; Hyderabad, Telangana is a preferred India location. Redrob engagement: response rate 0.91, notice 60 days, behavior multiplier 0.93.
2	CAND_0000060	0.390	Strong fit	8.0-year Cloud Engineer currently at Razorpay; relevant skills include FastAPI, Kafka, LoRA; Coimbatore, Tamil Nadu is India-based; minor concern: response rate is only 0.22. Recruiter interaction profile: response rate 0.22, notice 120 days, behavior multiplier 0.76.
3	CAND_0000005	0.368	Strong fit	Accountant with 11.0 years at Stark Industries; relevant skills include Apache Flink; Gurgaon, Haryana is a preferred India location; minor concern: current title is not a strong match for the role. Platform activity signals: response rate 0.37, notice 30 days, behavior multiplier 0.77.
4	CAND_0000048	0.360	Short list	9.7-year Mobile Developer currently at CRED; Hyderabad, Telangana is a preferred India location; JD fit score components: production 0.21, core skills 0.22. Redrob engagement: response rate 0.65, notice 120 days, behavior multiplier 0.73.
	CAND_0000000		Short list	Currently a Civil Engineer at Wipro with 5.5 years; relevant skills include Spark, Gurgaon, Haryana is a preferred India location; minor concern: current

Top candidate

#1 CAND_0000031 (score 1.000) — Currently a Recommendation Systems Engineer at Swiggy with 6.0 years; aligns with the JD requirement for shipped retrieval/ranking systems; relevant skills include Embeddings, Pinecone, scikit-learn; Hyderabad, Telangana is a preferred India location. Redrob engagement: response rate 0.91, notice 60 days, behavior multiplier 0.93.

Download full ranked CSV

redrob_r...csv 4.6 KB

Submission

- ✓ GitHub repo — clean, complete, working code
- ✓ Ranked output — submission.csv (candidate_id, rank, score, reasoning)
- ✓ Methodology — METHODOLOGY.md + this deck
- ✓ Live demos — Render dashboard + HuggingFace Space

Reproduce: `python rank.py --candidates ./candidates.jsonl --out ./submission.csv`